INCH-POUND
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DETAIL SPECIFICATION

ROPE, NYLON (SOLID YARN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and size of spun yarn nylon rope.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

COMMERCIAL ITEM DESCRIPTION

A-A-59622 - Tags, Shipping and stock

(Copies of these documents are available online at https://assist.dla.mil).

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this documents are available online at https://www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

SAE INTERNATIONAL (SAE)

SAE AMS-STD-595 Colors Used in Government Procurement

(Copies of these documents are available from www.sae.org or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Government and contractor purchases. The requirements specified in 3.9 and 3.10 apply only to rope purchased directly by the Government. All other requirements apply both to rope purchased by a contractor as a component for an end item and to rope purchased directly by the Government.

3.3 Standard sample. The rope shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6. 4).

3.4 Material. The rope shall be fabricated from spun staple nylon yarn in a denier to meet the requirements of this specification.

3.5 Construction. The rope shall be of three strand construction to conform to the requirements specified herein. The lay of the rope shall be right-hand or ”Z” lay. The length of ten turns of the finished rope shall be not less than 7 inches nor more than 8 inches when tested as specified in 4.3.4.

3.6 Physical requirements.

3.6.1 Circumference. The circumference of the rope shall be .750 ± .125 inch when tested as specified in 4.3.4.

3.6.2 Length per pound. The length per pound of the rope shall be not less than 47 feet when tested as specified in 4.3.4.

3.6.3 Breaking strength. The rope shall have a minimum breaking strength of not less than 850 pounds, and the loss in breaking strength after heat aging of the rope shall not exceed 10 percent when tested as specified in 4.3.4.

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3.6.4 Elongation. The elongation of the rope shall be no more than 35 percent when tested as specified in 4.3.4.

3.7 Color. Unless otherwise specified (see 6.2), the rope shall be dyed Green 383 Camo, to approximate color chip 34094 of SAE AMS-STD-595 (see 6.6).

3.7.1 Colorfastness. The dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.3.4. When no standard sample has been established, the dyed rope shall show “good” fastness to weathering when tested as specified in 4.3.4.

3.7.2 Matching. The color of the finished rope shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.8 Extraneous material. No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall be no more than 4.0 percent when tested as specified in 4.3.4.

3.9 Identification. Each spool (reel) of rope shall have a ticket (identification tag) attached for identification purposes. The ticket shall be in accordance with A-A-59622. All ticket entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket entries shall contain the following information:

National stock number
Nomenclature
Specification number
Length
Size (circumference)
Contract number and date
Contractor’s name
Date of manufacture (month and year)

3.10 Put-up. Unless otherwise specified (see 6.2), the rope shall be put-up in nonreturnable (reels) in the minimum 600-foot lengths. A plus tolerance of 10 percent is allowed on the length of any spool (reel) specified. The rope, when put-up in spools (reels), shall be in one continuous piece and so wound that each turn and layer is free from entanglement. The ends of all rope shall be cut off squarely and securely whipped or heat sealed to prevent fraying or untwisting.

3.11 Workmanship. The rope shall conform to the quality of product established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.3.2 and 4.3.3 and tested for the characteristics specified in 4.3.4.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ-Z1.4.

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4.3.1 Component and material inspections. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the material conforms to the requirements specified in 3.4.

4.3.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of spools (reels). The sample unit shall be one spool (reel), but not less than 100 feet. The inspection level shall be I.

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Table I. End item visual defects.

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	101	
	Chafed or damaged	102	
	Kink	103	
	Broken or loose end	104	
	Bulged strand	105	
	Darting yarn 1/	106	
	Strand knot	107	
Cleanness	Spot or stain, clearly visible 2/	108	
Identification	Omitted, incorrect, illegible, or not as specified	109	
	Insecurely attached		201

1/ Darting yarns are internal yarns which project through the cover yarns of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.3.3 End item length and winding examination. The rope shall be examined for the defects listed below. The lot size shall be expressed in units of spools (reels) of rope. The sample unit shall be one spool (reel) of rope. The inspections level shall be S-3. For lots consisting of 500 or fewer units, the sample size shall be 10.

Defects

Total length of rope on any spool is less than or greater than the length specified (see 3.10).

Length of rope on spool (reel) is less than or greater than the length marked on ticket (identification tag).

Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of rope.

Any end frayed or not cut off squarely or neatly.

Any end not securely whipped or heat sealed.

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4.3.4 End item testing. The end items shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in table II, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes specified in the applicable test methods. The lot size shall be expressed in units of spools (reels) of rope. The sample unit shall be one spool (reel) of rope. The sample size shall be in accordance with table III. The lot shall be unacceptable if one or more units fail to meet any requirement. All test reports shall contain the individual values utilized in expressing the final results. Tests to determine compliance with document requirements may be made under prevailing atmospheric conditions. In case of dispute, tests shall be made upon material that has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE II. End Item tests.

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Number of strands	3.5	Visual	1	Pass or fail
Lay of rope	3.5	4050	-	-
Length per 10 turns	3.5	6000	-	-
Circumference	3.6.1	6003	-	-
Length per pound	3.6.2	6004	-	-
Breaking strength:				
Initial	3.6.3	6015.1	-	-
After heat aging	3.6.3	4.4.1	2	Average of 2 determinations to nearest 10 pounds calculated to 1 percent
Elongation	3.6.4	6015.1	-	-
Colorfastness: to weathering	3.7.1	5671.1 1/	-	-
Shade matching	3.7.2	3.7.2	1	Pass or fail
Finish (extraneous material)	3.8	2/	-	-
Extractable matter	3.8	2611	-	-

1/ The specimens shall be exposed for 40 standard fading hours.

2/ Unless otherwise specified, a supplier's certificate of compliance shall be furnished and will be acceptable for the stated requirement.

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TABLE V. Sample size.

Lot size spools (reels)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,000 and over	5

4.4 Methods of inspection.

4.5.1 Breaking strength after aging. The breaking strength specimens, prepared as specified in Method 6015 of FED-STD-191, shall be heated for 5 days in a convection air oven at 175° ± 2°F. After aging, the specimens shall be allowed to reach equilibrium under standard conditions before being tested for strength. Breaking strength shall be determined in accordance with Method 6015. The percent change in strength shall be calculated based on the average strengths initially and after heat aging.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rope is intended to be used for lacing pontoon floats.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.1, 4.2 and 6.3).
c. Put-up, if other than specified (see 3.10).
d. Packaging requirements (see 5.1).
e. Color, if other than Green 383 Camo (see 3.7).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Lacing
Pontoon
Spools

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6.6 Color. Olive green and olive drab colors have been replaced by Camouflage Green 383. Any item which previously required any olive green or olive drab color will now require Camouflage Green 383.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SH
DLA - CQ

Review activities:
Army - CR4, MD

Preparing Activity:
DLA - CQ

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